DETECTION OF EVENTS CAUSING PLUGGAGE OF A COAL-FIRED BOILER: A DATA MINING APPROACH

Paper Summary

SUMMARY PASSAGE 1

Introduction

This study concentrates on an ash fouling condition that causes boiler shutdowns (several times a year) on a 750 MW tangential coal-fired boiler. The ash fouling leads to pluggage in the reheater section of the boiler. Once the build up becomes substantial the boiler performance is negatively affected.

SUMMARY PASSAGE 2

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The most effective method to segment the data is by determining=estimating the approximate date of failure and marking it as the last observation of the final time window. In this application the failure event was defined by the date when the boiler was derated due to the pluggage. The cause of the shutdown was confirmed through visual inspection of the affected region.

SUMMARY PASSAGE 3

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To identify clusters of data that potentially cause pluggage 45 different clusters per time window were analyzed. The TW4 clusters are especially important since the boiler pluggage was initiated during this time window.

SUMMARY PASSAGE 4

5: Analysis Of Knowledge And Validation

After the completion of steps 1 through 4 several potential adverse event clusters were identified and rules and knowledge have been extracted from the original data set. An additional three data sets were analyzed to further validate and isolate the specific events that lead to the boiler pluggage (Table 5). The validation approach is discussed in the next section.

SUMMARY PASSAGE 5

Statistical And Visual Analysis

There was no data for this parameter from confirmation data set 1, so it was ignored in this analysis. It is evident that confirmation data set 2 (no pluggage) is always lower than the two data sets that exhibited the pluggage. The NO6 O2 PROBE M graph shown in Figure 5 exhibits a trend similar to that of the NO3 O2 PROBE M graph.

SUMMARY PASSAGE 6

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The analysis of the STACK FLOW M demonstrated an interesting trend (see Figure 7). The graph depicts the general relationship that lower values of stack flow are present in the data set without boiler pluggage.

SUMMARY PASSAGE 7

Analysis Of Pluggage Of A Coal-Fired Boiler

In each of the three data sets affected by the pluggage at least one of the seven coal feeders was turned off for at least one time window (approximately one week). For data set 2 all feeders were on throughout the six time windows. This relationship may indicate that the pluggage could be due to changes in boiler output or sudden demand shifts.

SUMMARY PASSAGE 8

Rule Validation

These results are even more significant given the fact that there were zero predictions for these two time windows in the second confirmation data set. These results indicate that the rules captured the unique relationships and events that lead to the boiler pluggage. These observations were extracted and analyzed to provide even more details of the events.

SUMMARY PASSAGE 9

Cluster Analysis

To validate the adverse event clusters derived from the clustering algorithm, the adverse event clusters were compared to clusters generated from confirmation data sets 1 and 2. Recall that boiler tube pluggage was present throughout the entire confirmation data set 1 and that boiler tube pluggage was initiated during TW4 of the original data set. Given this, it is hypothesized that the TW4 potential adverse event clusters from the original data set should be highly similar to all time window clusters of confirmation data set 1.

SUMMARY PASSAGE 10

Conclusions

A rule-based data mining, statistical, and clustering approach was utilized to identify events that lead to ash pluggage in the reheater section of an industrial coal-fired boiler. This approach increased the understanding and the potential causes of the pluggage and provided a knowledge base Table 10. Potential adverse event clusters from Table 3 Potential adverse event clusters Cluster no.